

Understanding Data with Visualization



CONTENTS

- Introduction
- Why visualization matters?
- Univariate Plots: Understanding Attributes Independently
- Bivariate Plots
- Multivariate Plots: Interaction Among Multiple Variables
- When to use what plot?

INTRODUCTION

Data visualization is a way to show numbers or information as pictures like charts and graphs that are easier for human brain to understand.

This can include a variety of visual tools such as:

- **Charts:** Bar charts, line charts, pie charts, etc.
- **Graphs:** Scatter plots, histograms, etc.
- **Maps:** heat maps, etc.
- **Dashboards:** Interactive platforms that combine multiple visualizations.

WHY VISUALIZATION MATTERS?

- For understanding the data clearly
- Decision making
- Finding solution to a problem
- To find relationship among the data
- Comparative analysis
- Data visualization save time
- Represents data beautifully
- It tells a story
- Provides perspective on the data

UNIVARIATE PLOT

- A univariate plot is a type of graph that shows data for just **one variable**.
- “Uni” means one, so it focuses on a single column or feature in a dataset.
- It helps us to understand the distribution, shape, and spread of that one variable.
- Common univariate plots include **histograms, box plots, and density plots**.
- Univariate plots are useful for **spotting patterns like skewness or unusual values**.
- They are often the first step in exploring data during analysis.

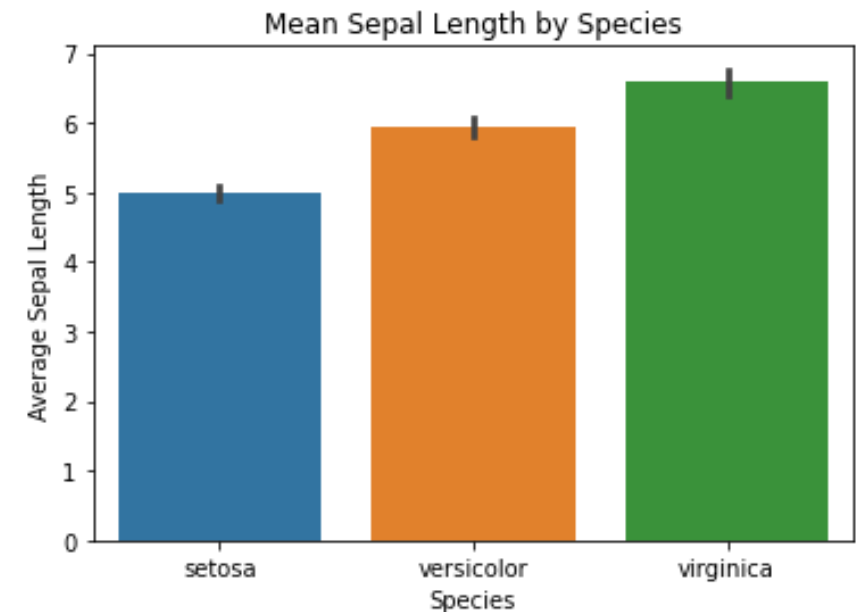
UNIVARIATE PLOT FOR CATEGORICAL DATA

- Univariate plots for **categorical data** focus on showing the **frequency** (or count) of each category in a single variable.
- It is used to visualize the distribution of a single categorical variable.
- Since categories are not numeric, we cannot use histograms or density plots—**instead, we use bar plot, count plots and pie charts.**
- These plots help answer questions like:
 - How many samples belong to each category?
 - Are some categories more common than others?
 - Are the categories balanced?

BAR PLOT

A bar plot displays the frequency or proportion of each category in a categorical variable. Each bar represents a category, with the height of the bar representing the frequency or proportion of observations in that category.

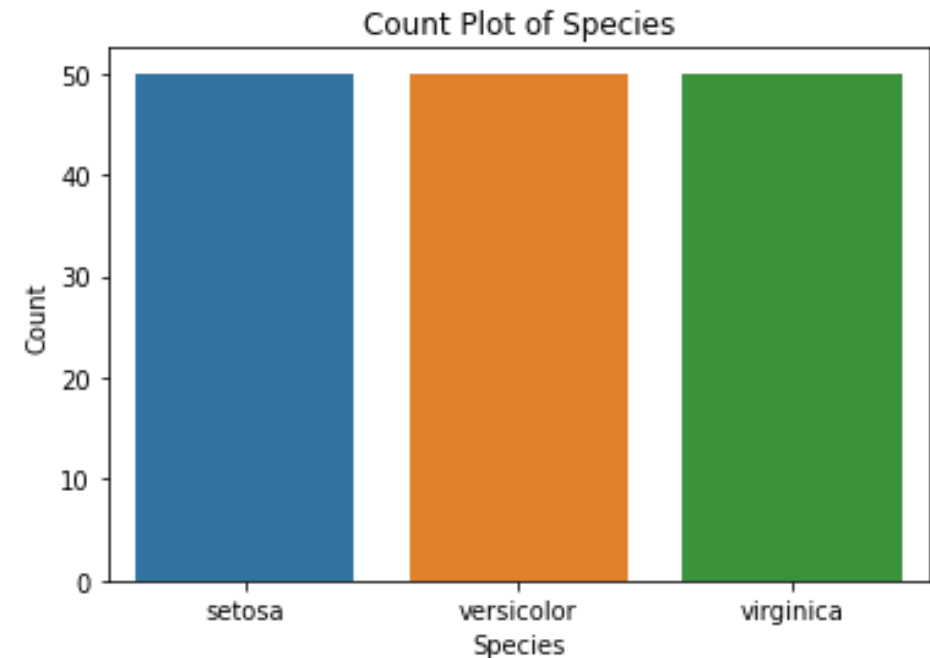
```
# Bar Plot - Mean Sepal Length per Species
plt.figure(figsize=(6, 4))
sns.barplot(x='species', y='sepalLength', data=df, estimator=np.mean)
plt.title("Mean Sepal Length by Species")
plt.xlabel("Species")
plt.ylabel("Average Sepal Length")
plt.show()
```



COUNT PLOT

A count plot is a type of bar plot that displays the count of observations in each category. It is similar to a bar plot, but instead of displaying the frequency or proportion, it **displays the actual count of observations**.

```
# Categorical variable: 'species'  
# Count Plot  
plt.figure(figsize=(6, 4))  
sns.countplot(x='species', data=df)  
plt.title("Count Plot of Species")  
plt.xlabel("Species")  
plt.ylabel("Count")  
plt.show()
```

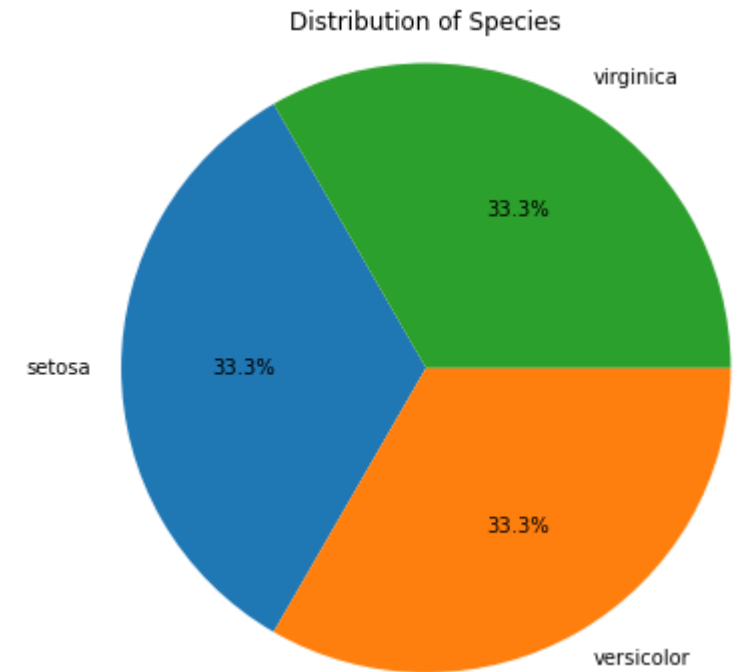


PIE CHART

A pie chart displays the **proportion of each category** in a categorical variable. Each slice of the pie represents a category, with the angle of the slice representing the proportion of observations in that category.

```
# Count the number of each species
species_counts = df['species'].value_counts()

# Plot pie chart
plt.figure(figsize=(6, 6))
plt.pie(species_counts, labels=species_counts.index, autopct='%1.1f%%', startangle=120)
plt.title("Distribution of Species")
# Equal aspect ratio makes the pie chart circular.
plt.axis('equal')
plt.show()
```



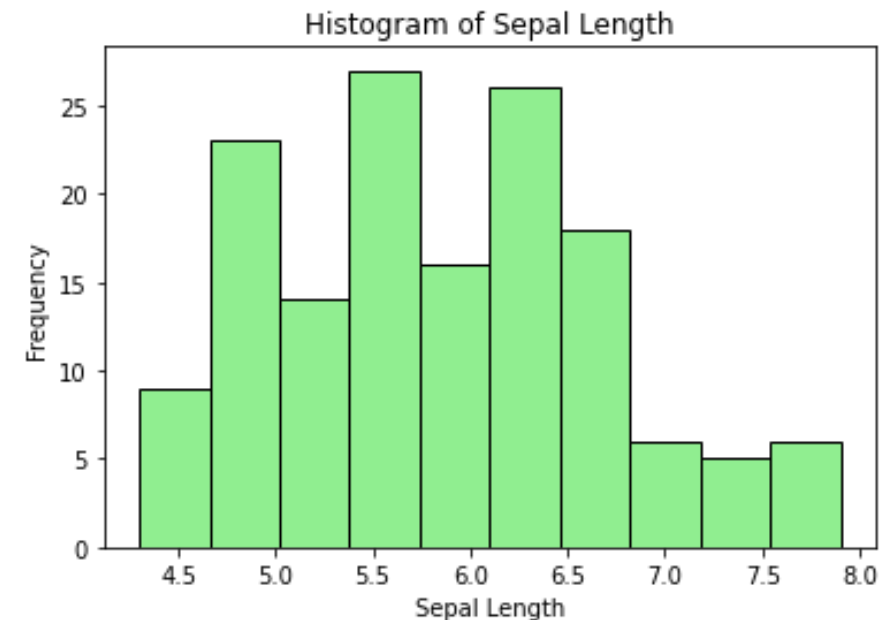
UNIVARIATE PLOT FOR NUMERICAL DATA

- Univariate plots for numerical data are used to visualize the distribution of a single numerical variable.
- They are useful for identifying **patterns, outliers, and skewness** in the data.
- Common univariate plots for numerical data include **histograms, density plots, and box plots**.
- Numerical univariate plots help us understand the **shape, center, and spread** of a single variable.
- We can see where most values lie and how concentrated or scattered, they are.
- These plots are key for spotting patterns and preparing data for analysis.

HISTOGRAMS

A histogram is a bar chart that displays the frequency of a numerical variable's values. It is created by dividing the data into intervals, called bins, and counting the number of observations that fall within each bin.

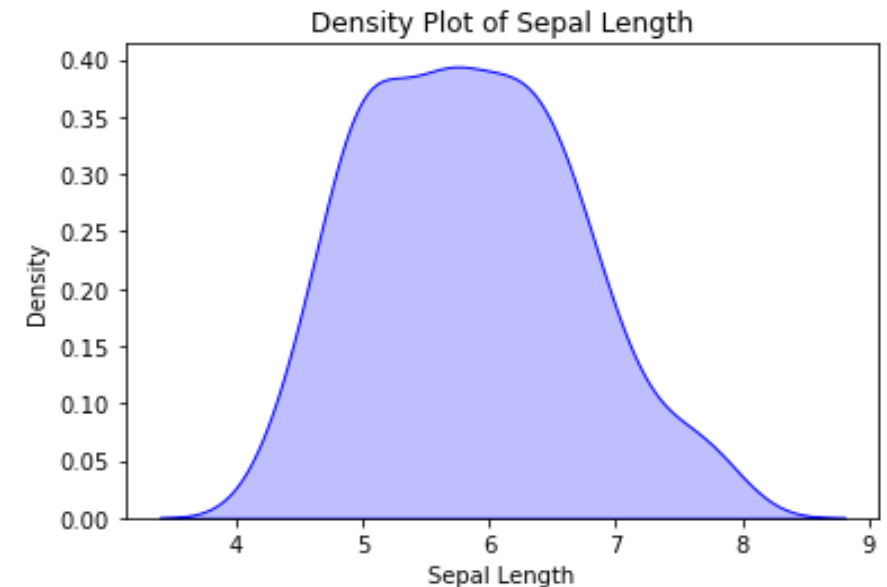
```
plt.figure(figsize=(6, 4))  
plt.hist(df['sepalLength'], bins=10, color='lightgreen', edgecolor='black')  
plt.title("Histogram of Sepal Length")  
plt.xlabel("Sepal Length")  
plt.ylabel("Frequency")  
plt.show()
```



DENSITY PLOT

A density plot is a smoothed version of a histogram. It uses a kernel density estimation (KDE) to estimate the probability density function of the data. Density plots are useful for comparing the distribution of two or more datasets.

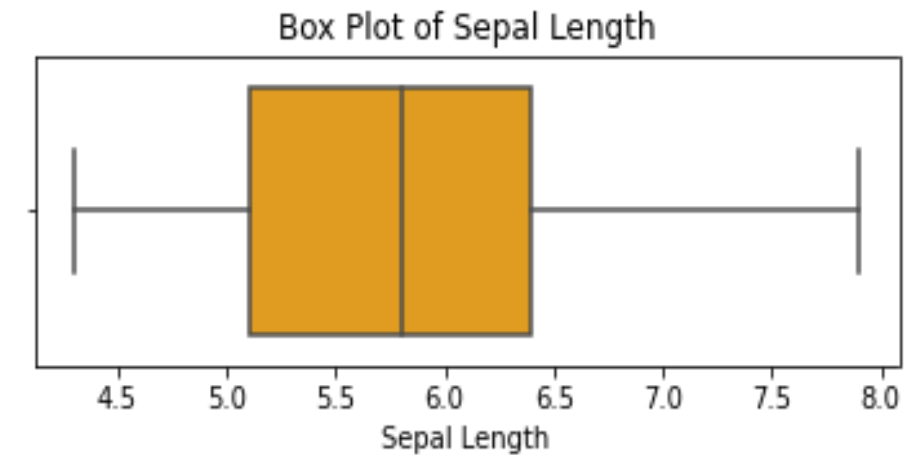
```
plt.figure(figsize=(6, 4))  
sns.kdeplot(df['sepalLength'], shade=True, color='blue')  
plt.title("Density Plot of Sepal Length")  
plt.xlabel("Sepal Length")  
plt.ylabel("Density")  
plt.show()
```



BOX PLOT

A box plot, also known as a **whisker plot**, displays the distribution of a numerical variable using five number summaries: **the minimum, first quartile, median, third quartile, and maximum**. It also shows any potential outliers.

```
plt.figure(figsize=(6, 2))  
sns.boxplot(x=df['sepalLength'], color='orange')  
plt.title("Box Plot of Sepal Length")  
plt.xlabel("Sepal Length")  
plt.show()
```



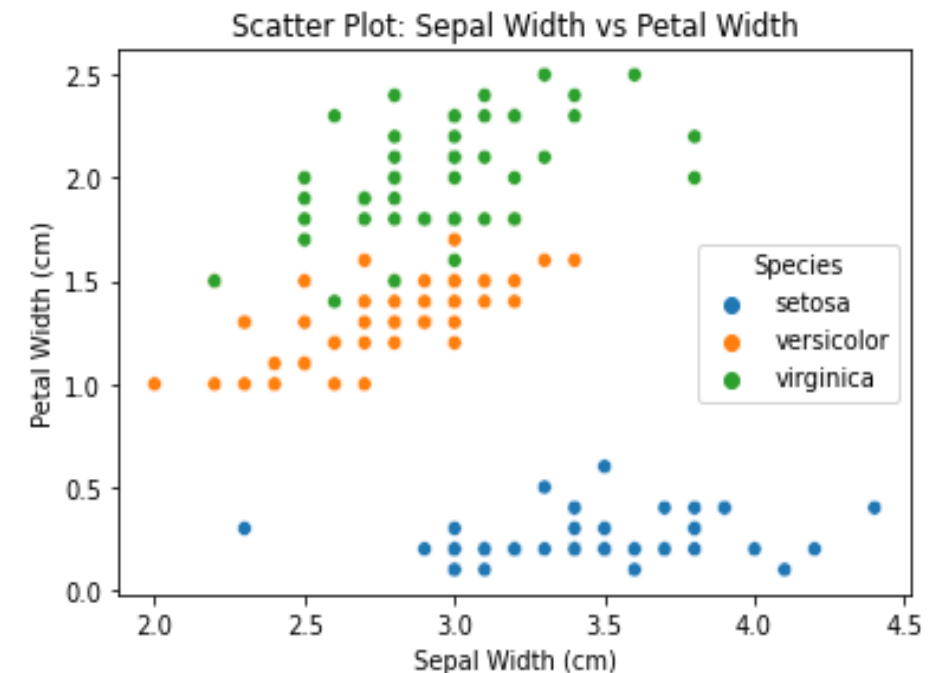
BIVARIATE PLOT

- A **bivariate plot** is a graph that shows the relationship between **two variables**.
- Common bivariate plots include **scatter plots, line plots, and bar plots**.
- Unlike univariate plots (which explore one variable), bivariate plots help answer:
 - Does one variable depend on another?
 - Is there correlation or pattern?
 - How do two variables interact?
- There are the main types of bivariate plot are used
 - Categorical Vs Numerical
 - Numerical Vs Numerical
 - Categorical Vs Categorical

SCATTER PLOTS

A scatter plot displays the relationship between two continuous variables. Each point on the plot represents an observation, with the x-axis representing one variable and the y-axis representing the other.

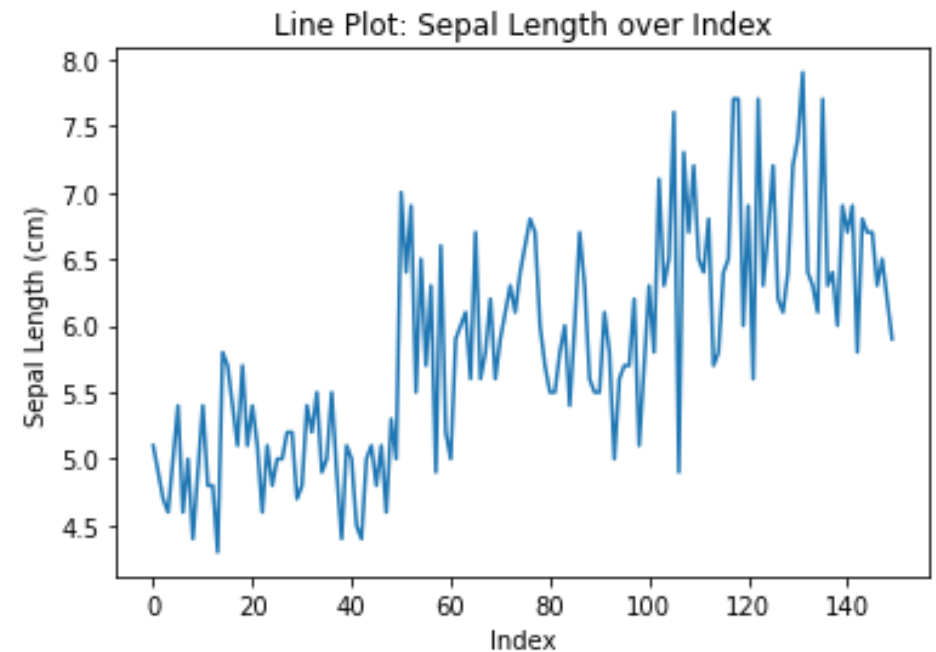
```
# Scatter plot of Sepal Width vs Petal Width
sns.scatterplot(x='sepalWidth', y='petalWidth', data=df, hue='species')
plt.title("Scatter Plot: Sepal Width vs Petal Width")
plt.xlabel("Sepal Width (cm)")
plt.ylabel("Petal Width (cm)")
plt.legend(title='Species', loc='center right')
plt.show()
```



LINE PLOT

A line plot displays the relationship between two continuous variables, with one variable represented on the x-axis and the other on the y-axis. Line plots are useful for showing trends over time.

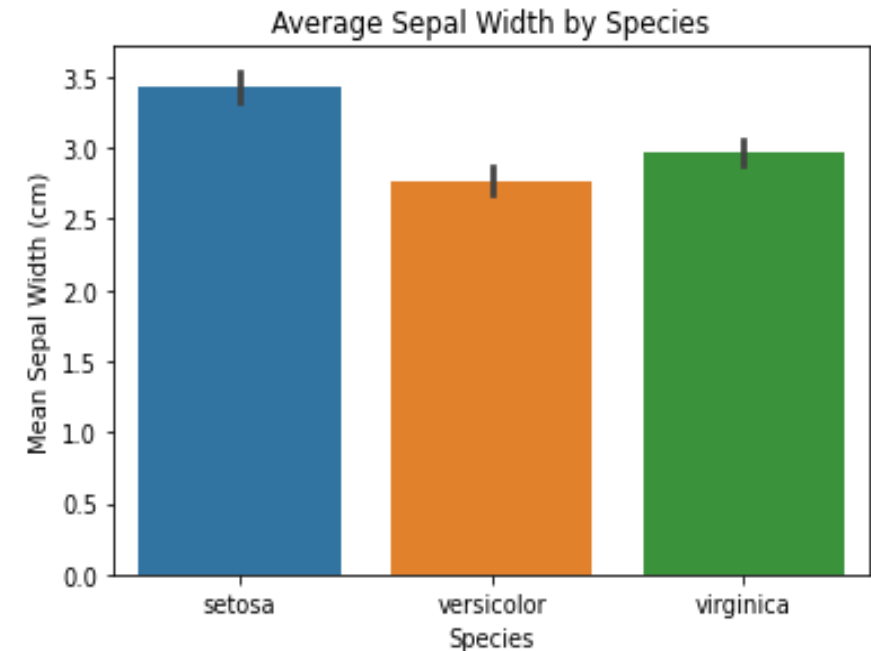
```
# Line plot of Sepal Length over index  
sns.lineplot(x=df.index, y='sepalLength', data=df)  
plt.title("Line Plot: Sepal Length over Index")  
plt.xlabel("Index")  
plt.ylabel("Sepal Length (cm)")  
plt.show()
```



BAR PLOT

A bar plot displays the relationship between two categorical variables. Each bar represents a category, with the height of the bar representing the value of the second variable.

```
# Bar plot showing mean Sepal Width for each species  
sns.barplot(x='species', y='sepalWidth', data=df, estimator=np.mean)  
plt.title("Average Sepal Width by Species")  
plt.xlabel("Species")  
plt.ylabel("Mean Sepal Width (cm)")  
plt.show()
```



CATEGORICAL VS NUMERICAL

Bivariate plots for numerical vs. categorical data are used to visualize the relationship between a numerical variable and a categorical variable. They are useful for identifying **differences, trends, and patterns**. Common bivariate plots for numerical vs. categorical data include **box plots, violin plots, and swarm plots**.



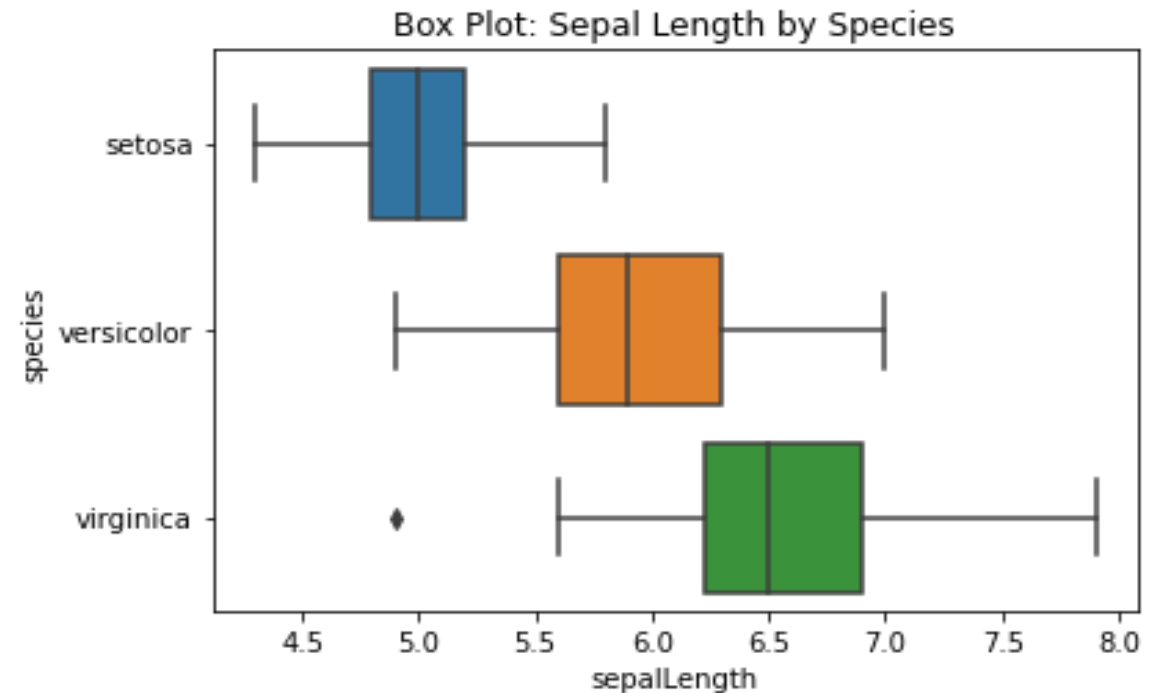
Categorical data



Numerical data

BOX PLOT

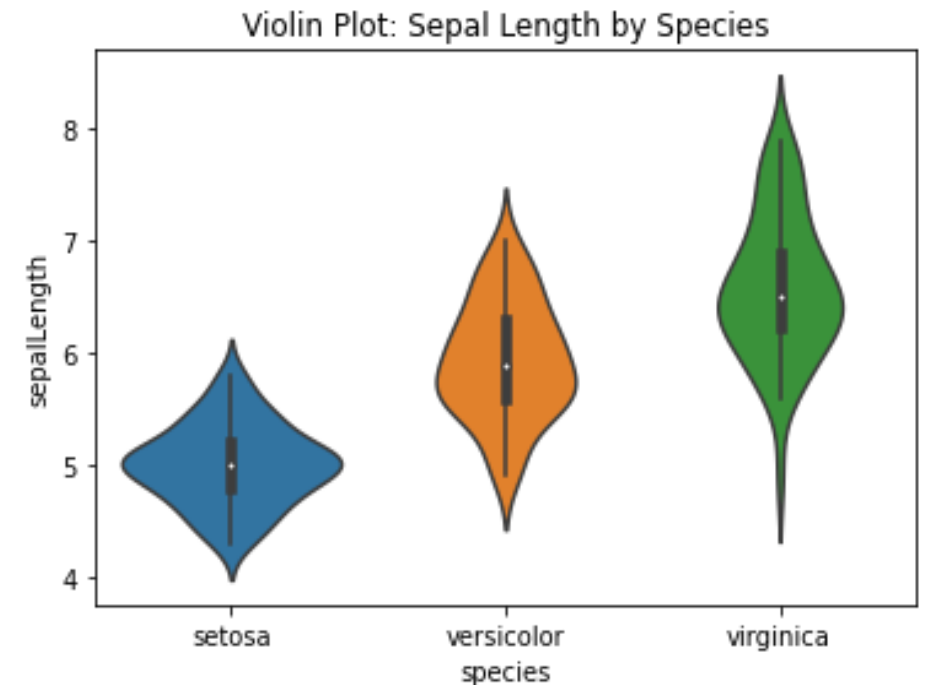
```
# Box Plot  
sns.boxplot(x='sepalLength', y='species', data=df)  
plt.title("Box Plot: Sepal Length by Species")  
plt.show()
```



VIOLIN PLOT

```
# Violin Plot  
sns.violinplot(x='species', y='sepalLength', data=df)  
plt.title("Violin Plot: Sepal Length by Species")  
plt.show()
```

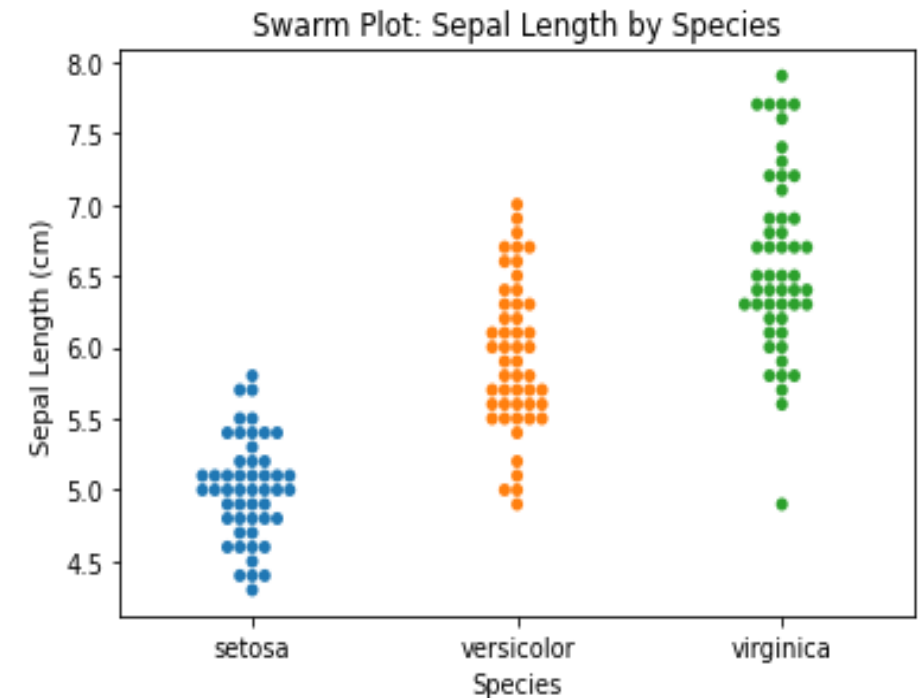
A violin plot displays the distribution of a numerical variable for each category in a categorical variable, similar to a box plot. However, instead of showing a box and whiskers, it shows a **kernel density estimate of the distribution**.



SWARM PLOT

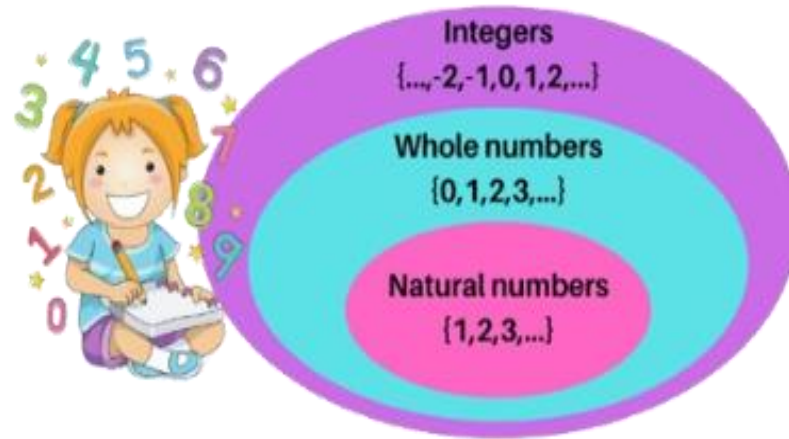
A swarm plot displays the individual observations of a numerical variable for each category in a categorical variable. Each point represents an observation, with the position along the x-axis representing the category and the position along the y-axis representing the value of the numerical variable.

```
# Swarm plot of Sepal Length for each species
sns.swarmplot(x='species', y='sepalLength', data=df)
plt.title("Swarm Plot: Sepal Length by Species")
plt.xlabel("Species")
plt.ylabel("Sepal Length (cm)")
plt.show()
```



NUMERICAL VS NUMERICAL

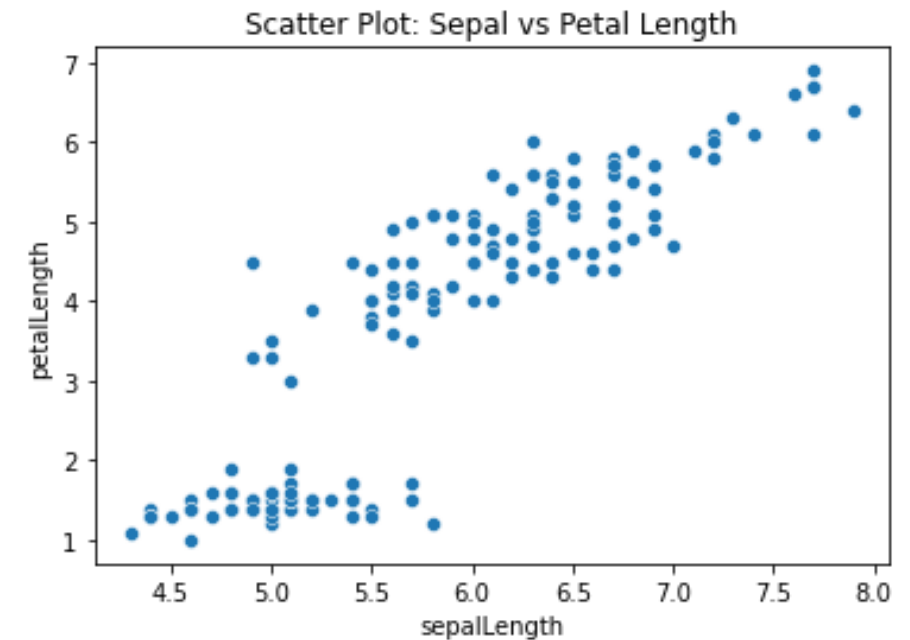
Bivariate plots for numerical vs. numerical data are used to visualize the relationship between two numerical variables. They are useful for **identifying trends, patterns, and correlations**. Common bivariate plots for numerical vs. numerical data include **scatter plots, line plots, and 2D histograms**.



SCATTER PLOT

A scatter plot displays the relationship between two numerical variables. Each point on the plot represents an observation, with the x-axis representing one variable and the y-axis representing the other.

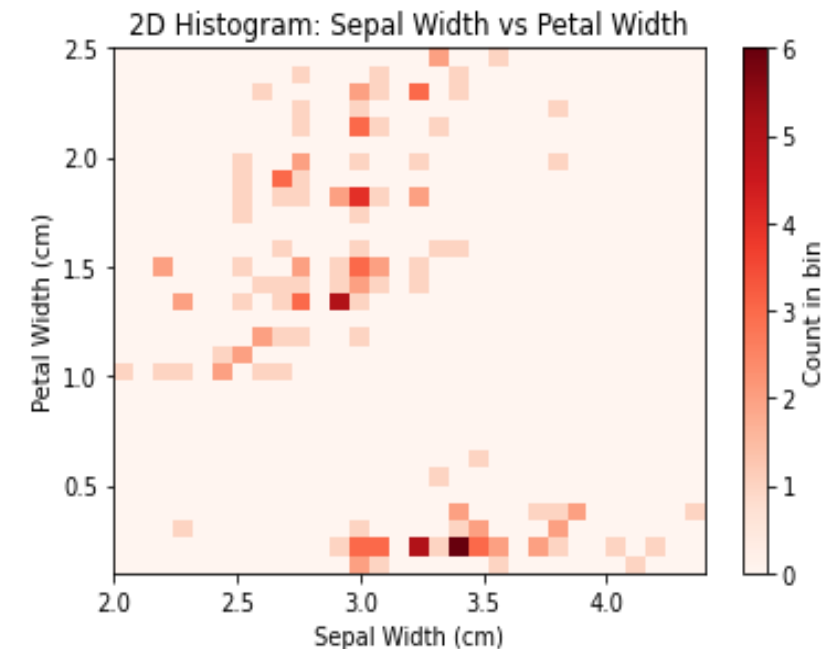
```
# Scatter Plot  
sns.scatterplot(x='sepalLength', y='petalLength', data=df)  
plt.title("Scatter Plot: Sepal vs Petal Length")  
plt.show()
```



2D HISTOGRAM

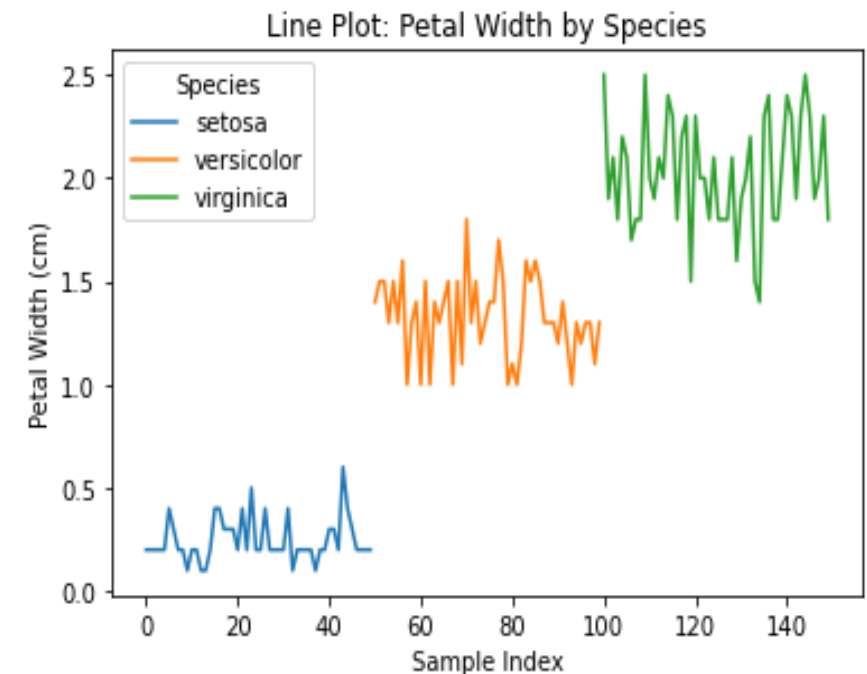
A 2D histogram displays the relationship between two numerical variables using a grid of bins. Each bin represents a range of values for both variables, with the color of the bin representing the number of observations that fall within that range.

```
plt.hist2d(df['sepalWidth'], df['petalWidth'], bins=30, cmap='Reds')  
plt.colorbar(label='Count in bin')  
plt.title("2D Histogram: Sepal Width vs Petal Width")  
plt.xlabel("Sepal Width (cm)")  
plt.ylabel("Petal Width (cm)")  
plt.show()
```



LINE PLOT

```
# Line plot of Petal Width for each species over sample index  
sns.lineplot(x=df.index, y='petalWidth', hue='species', data=df)  
plt.title("Line Plot: Petal Width by Species")  
plt.xlabel("Sample Index")  
plt.ylabel("Petal Width (cm)")  
plt.legend(title='Species')  
plt.show()
```



CATEGORICAL VS CATEGORICAL

Bivariate plots for categorical vs. categorical data are used to visualize the relationship between two categorical variables. They are useful for identifying patterns, frequencies, and associations. Common bivariate plots for categorical vs. categorical data include **contingency tables, mosaic plots, and stacked bar plots.**



Setosa



Versicolor



Virginica

CONTINGENCY TABLE

A contingency table displays the frequency of observations for each combination of categories in two categorical variables. It is a tabular representation of the joint distribution of the two variables.

```
# Add a new categorical column with 3 flower types
def assign_flower_type(species):
    if species == 'setosa':
        return 'Type A'
    elif species == 'versicolor':
        return 'Type B'
    else:
        return 'Type C'

df['flower_type'] = df['species'].apply(assign_flower_type)

# Contingency table between species and flower_type
contingency_table = pd.crosstab(df['species'], df['flower_type'])
print(contingency_table)
```

Output

flower_type	Type A	Type B	Type C
species			
setosa	50	0	0
versicolor	0	50	0
virginica	0	0	50

MULTIVARIATE PLOT

- Multivariate analysis is a **statistical technique** that involves analyzing multiple variables simultaneously.
- It is used to **identify patterns, relationships, and structures** in complex datasets.
- Multivariate analysis can be used for various purposes, such as **data reduction, classification, and prediction.**
- One or more variables can be **numerical** or **categorical.**
- It provides a richer view than univariate or bivariate plots.
- Example of multivariate plots are **scatter matrix plots, correlation matrix plot, pair plot, 3D plot** etc.

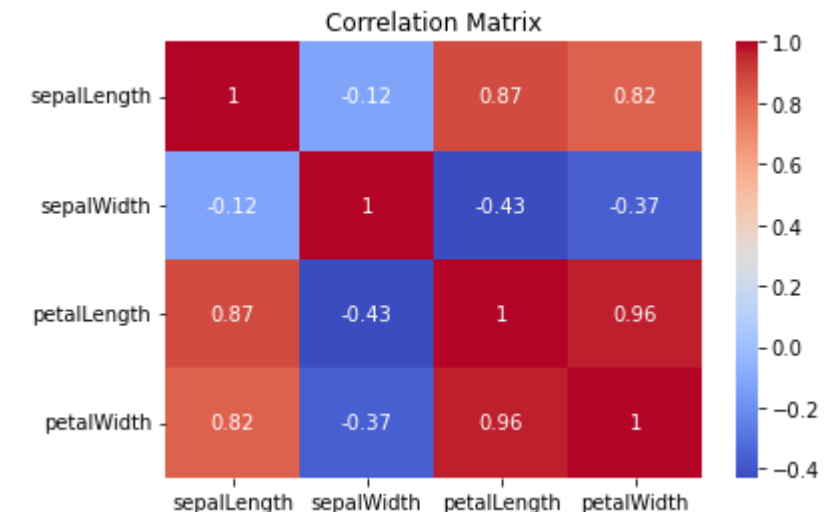
HEAT MAP

We use a **heatmap** to see how strongly different columns in the dataset are related. It uses **colors** to show the strength of the relationship. Correlation values range from **-1 to 1**:

- **+1** means a strong positive relationship
- **-1** means a strong negative relationship
- **0** means no relationship

```
# Compute correlation matrix
corr = df[['sepalLength', 'sepalWidth', 'petalLength', 'petalWidth']].corr()

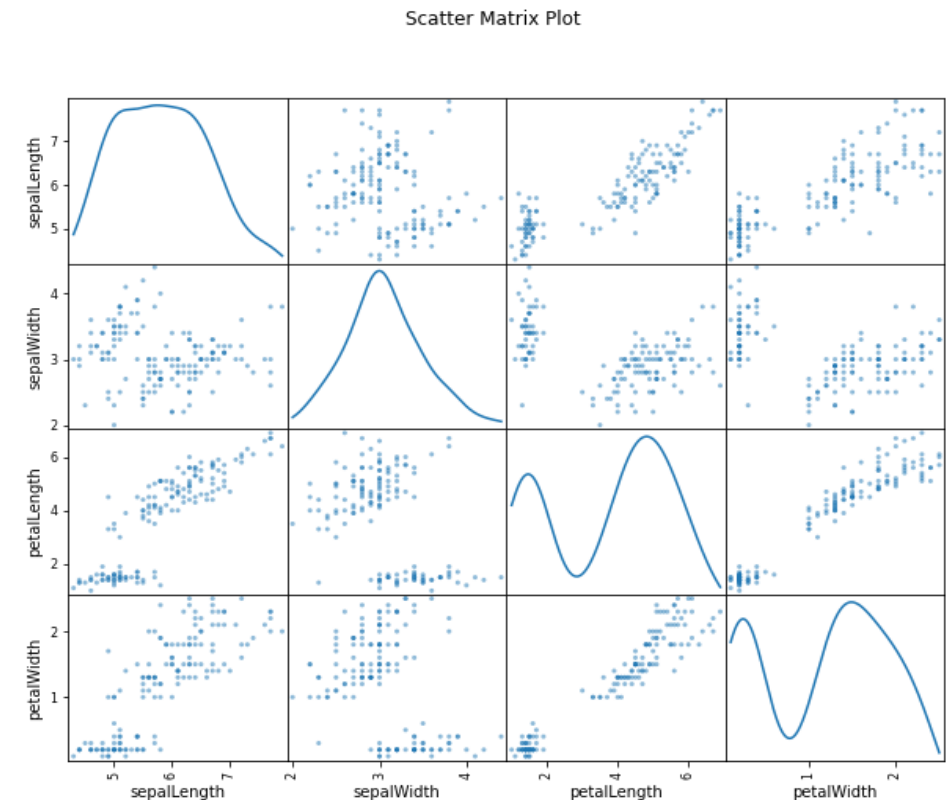
# Plot heatmap
sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.title("Correlation Matrix")
plt.show()
```



SCATTER MATRIX PLOT

A **scatter matrix plot** (also called a **pair plot**) is a grid of scatter plots that shows the **relationships between multiple numerical variables** in a dataset — all at once.

```
from pandas.plotting import scatter_matrix  
  
scatter_matrix(df[['sepalLength', 'sepalWidth', 'petalLength', 'petalWidth']],  
              figsize=(10, 8), diagonal='kde')  
plt.suptitle("Scatter Matrix Plot")  
plt.show()
```



WHEN TO USE WHAT PLOT?

Plot Type	Use Case	Data Type	Insights Gained
Histogram	View distribution of a single variable	Numerical (Univariate)	Shape, spread, skew
Box Plot	Compare distributions across groups	Categorical vs Numerical	Median, IQR, outliers
Bar Plot	Compare averages or counts between categories	Categorical vs Numerical	Count or mean comparison
Pie Chart	Show part-to-whole proportions	Categorical	Relative size of categories
Swarm/Strip Plot	Show all values per category	Categorical vs Numerical	Spread and outliers
Scatter Plot	Show relation between two numerical variables	Numerical vs Numerical	Trend, correlation, clusters
Line Plot	Track changes over time/index	Numerical (Time/Index-based)	Trend and fluctuations

WHEN TO USE WHAT PLOT?

Plot Type	Use Case	Data Type	Insights Gained
2D Histogram/Hexbin	Show density where points overlap	Numerical vs Numerical	Dense regions and data spread
Stacked Bar Plot	Show subcategory counts by group	Categorical vs Categorical	Part-to-whole comparisons
Mosaic Plot	Visualize proportions across 2+ categorical variables	Categorical vs Categorical	Grouped frequency or ratio patterns
Pair Plot / Scatter Matrix	Visualize pairwise numeric relationships	Multiple Numerical	Distribution, correlation, clustering
Correlation Heatmap	Show how strongly numerical features are related	Multiple Numerical	Positive/negative correlation strengths
3D Scatter Plot	Explore 3 numerical variables in 3D space	3 Numerical Variables	Complex spatial trends and clusters

THANK YOU